



**ACADEMIC CENTRAL**

PRACTICE EXAMINATION

**GRADE 10**

**CIVIL TECHNOLOGY**

**MARKING GUIDELINE**

**MARKS: 100**

**Marking guideline for the Academic Central Grade 10 Civil Technology practice examination.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark. Accept other relevant, correct answers. Penalise once per question for missing units.

## QUESTION 1

**1**      1.1 B ✓✓   1.2 B ✓✓   1.3 A ✓✓   1.4 B ✓✓   1.5 B ✓✓      [10]

## QUESTION 2

- |            |                                                                                                                                                                                                              |     |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>2.1</b> | Any THREE: provide a safe workplace ✓; provide protective equipment ✓; train workers in safe work methods ✓; maintain machines and equipment; report and investigate accidents; provide first-aid facilities | (3) |
| <b>2.2</b> | (a) safety goggles / face shield (and ear protection) ✓ (b) dust mask and gloves ✓ (c) safety harness ✓                                                                                                      | (3) |
| <b>2.3</b> | Red: prohibition (do not) / fire equipment ✓; blue: mandatory (must do) ✓; yellow: warning / caution ✓; green: safe condition / first aid / emergency exit ✓                                                 | (4) |
| <b>2.4</b> | Wear gloves; press a clean cloth or dressing firmly on the wound to stop the bleeding ✓; raise the hand above the heart ✓; bandage it and get medical help (call for help if the bleeding does not stop) ✓.  | (3) |
| <b>2.5</b> | To remove dust, fumes and harmful gases ✓ so that workers can breathe clean air and to reduce the risk of fire ✓.                                                                                            | (2) |

**[15]**

### QUESTION 3

- |            |                                                                                                                                                                         |     |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| <b>3.1</b> | Total parts = 1 + 2 + 3 = 6 ✓; cement = $0,6 \times \frac{1}{6} = 0,1 \text{ m}^3$ ✓; sand = 0,2 m <sup>3</sup> ✓; stone = 0,3 m <sup>3</sup> ✓                         | (4) |
| <b>3.2</b> | Keeping concrete moist (and at a suitable temperature) for several days after it is placed ✓✓, so that it can harden properly, reach its full strength and not crack ✓. | (3) |
| <b>3.3</b> | Any TWO: easy to work with ✓; strong for its weight ✓; renewable; good insulator; attractive appearance                                                                 | (2) |
| <b>3.4</b> | Any THREE: knots ✓; warping (bowing, cupping, twisting) ✓; shakes / splits ✓; rot; insect damage                                                                        | (3) |
| <b>3.5</b> | Any TWO: face brick ✓; common (stock) brick ✓; cement (concrete) brick; engineering brick                                                                               | (2) |
| <b>3.6</b> | It prevents moisture from rising up from the ground into the wall ✓✓ (rising damp).                                                                                     | (2) |
| <b>3.7</b> | Any TWO: paint or varnish ✓; treat with a preservative such as creosote or by pressure treatment ✓; keep dry                                                            | (2) |
| <b>3.8</b> | Concrete is strong in compression but weak in tension ✓; steel gives the concrete tensile strength so it does not crack under load ✓.                                   | (2) |

**QUESTION 4**

- 4.1** Claw hammer: drive and pull out nails ✓; tenon saw: make fine, straight cuts in wood (joints) ✓; brick trowel: spread mortar when laying bricks ✓; cold chisel: cut or chip metal, brick or concrete ✓; builder's line: keep brick courses straight and level ✓ (5)
- 4.2** Any THREE: check cables and plugs for damage before use ✓; unplug before changing blades or bits ✓; clean after use ✓; store in a dry place; have them serviced; use only for their intended purpose (3)
- 4.3** Place the level on a flat surface and read the bubble ✓; turn it end for end (180°) in the same position ✓; if the bubble is in the same place both times, the level is accurate ✓. (3)
- 4.4** Any TWO, each with a use: mortise and tenon ✓, used in door and window frames, tables ✓; dovetail ✓, used in drawers and boxes ✓; butt joint, used for simple frames; halving joint, used for frames (4)

[15]

**QUESTION 5**

- 5.1** To spread the load of the building evenly onto the soil ✓ so that the building does not sink or crack ✓. (2)
- 5.2**  $V = 12 \times 0,6 \times 0,2 \checkmark\checkmark = 1,44 \text{ m}^3 \checkmark$  (3)
- 5.3.1**  $\text{Area} = 6 \times 2,4 \checkmark = 14,4 \text{ m}^2 \checkmark$ ;  $14,4 \times 55 \checkmark = 792 \text{ bricks } \checkmark$  (4)
- 5.3.2**  $792 \times 1,05 = 831,6 \checkmark \rightarrow 832 \text{ bricks } \checkmark$  (2)
- 5.3.3**  $832 \times R4,50 \checkmark = R3\,744 \checkmark$  (2)
- 5.4** Any FOUR: rafters ✓; purlins / battens ✓; wall plate ✓; ridge ✓; tie beam; king post; fascia board (4)
- 5.5** To carry the load of the brickwork above a door or window opening ✓✓. (2)
- 5.6** Any THREE: makes the soil stronger so it can carry the load ✓; prevents settlement and cracks ✓; removes air pockets ✓; reduces water penetration (3)
- 5.7** Measure 3 m along one line ✓ and 4 m along the other line ✓; adjust until the diagonal between the two points is exactly 5 m, then the angle is 90° ✓. (3)

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**QUESTION 6**

- 6.1** (a) visible outlines / edges ✓✓ (b) hidden detail (edges that cannot be seen) ✓✓ (c) centre lines / lines of symmetry ✓✓ (6)
- 6.2** DPC: damp-proof course ✓; FFL: finished floor level ✓; NGL: natural ground level ✓ (3)
- 6.3**  $4\,000 \text{ mm} \div 50 \checkmark = 80 \text{ mm } \checkmark$  (2)
- 6.4** Front view (front elevation) ✓; top view (plan) ✓; side view (side / end elevation) ✓ (3)
- 6.5** To draw large objects on paper in correct proportion ✓. (1)

[15]

TOTAL: 100

END OF MARKING GUIDELINE